

GATE-KD: ALIGNING TEACHER TARGETS TO THE STUDENT SPACE

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ABSTRACT

Distilling text embeddings across architectures involves two mismatches: teacher and student may have different widths and different coordinate systems. Projection resolves the first, but leaves the second: rotating every projected target preserves all pairwise similarities while changing the gradients of a pointwise loss. GATE-KD turns this ambiguity into an interface. It projects cached teacher endpoints, uses a corpus-level Procrustes fit to express them in coordinates aligned with the pretrained student, and supervises the student’s deployed embedding directly, without a learned cross-space map or teacher hidden states. Under a label-free, task-adapted protocol, the full recipe obtains the highest numerical average in three teacher–student configurations, with the clearest gain for a 22M student (+1.59 points). In a matched endpoint-only test, the fixed interface scores 74.28, versus 69.35 without alignment and 69.33 for the stronger learned linear map. Re-estimating the alignment every epoch adds only 0.16 points, while aligned random subspaces remain within 0.34 points of PCA despite retaining far less teacher energy. These results support student-aligned target presentation as the core design choice in the evaluated regime; an optional native-space H_0 term provides a smaller auxiliary gain.

1 INTRODUCTION

Text embeddings support retrieval, similarity, clustering, and classification, but large encoders can be costly to deploy. Knowledge distillation transfers their behavior to compact students using unlabeled text (Hinton et al., 2015; Reimers & Gurevych, 2020). Unlike class logits, embeddings from independently pretrained models need not share a width or a coordinate system. Prior methods bridge this mismatch with learned projectors, relational losses, or selected hidden layers (Park et al., 2019; Tian et al., 2020; Xu et al., 2023; Zhang et al., 2024a; Truong et al., 2025; Dao et al., 2026). We ask whether final teacher endpoints alone can provide a direct target for the embedding that the student will deploy.

Dimensional reduction and coordinate alignment solve different problems. Projection chooses which teacher variation survives at student width; the remaining axes can still be rotated. Imagine a map on a transparent sheet: rotating it changes no distance between cities, but changes every (x, y) coordinate. Likewise, an orthogonal rotation preserves all target similarities yet changes the gradients of pointwise distillation. We call choosing one such student-relative orientation *gauge fixing*. It selects a presentation for optimization, not canonical teacher coordinates.

This view leads to **GATE-KD**, Gauge-Aligned Target Embeddings for Knowledge Distillation. GATE-KD caches final teacher embeddings, projects them to student width, and uses closed-form orthogonal Procrustes alignment to orient the targets toward the pretrained student. Only the student is trained; the target map is fixed within each epoch and absent at inference. Procrustes is classical—the design choice is to apply it on the target side instead of learning a projector or resolving the orientation inside every minibatch loss. An optional native-space H_0 term transfers coarse connectivity scales.

Across three teacher–student pairs and nine sentence benchmarks, GATE-KD has the highest numerical average under a label-free, task-adapted protocol. The clearest margin is 1.59 points for the 22M student; the 0.38 and 0.48 margins for larger students are descriptive three-seed comparisons. In compact matched controls, the endpoint-only fixed interface exceeds the stronger of two learned

linear maps by 4.95 points. A fitted initial orientation is within 0.16 points of epoch-wise refitting, and aligned random subspaces are within 0.34 points of PCA despite much lower retained energy. Thus the main empirical story is the student-aligned presentation of a fixed endpoint target, rather than frequent alignment or PCA-specific variance retention.

We make three contributions:

- We separate cross-dimensional endpoint distillation into subspace selection and student-relative target orientation.
- We introduce a closed-form target-side alignment for direct endpoint supervision, with no trainable cross-space map or layer correspondence.
- We evaluate three model pairs and matched compact-student controls that isolate target construction, orientation, and refit frequency.

2 RELATED WORK

Prior work addresses three pieces of the problem: what representation to transfer, how to bridge different widths, and whether to compare coordinates or coordinate-free structure. We organize the literature along these axes.

Embedding and intermediate-layer distillation. Early text-embedding distillation directly regressed student sentence vectors onto teacher outputs (Reimers & Gurevych, 2020). Later methods augment endpoint transfer with contrastive, relative-geometric, multilingual, or task-specific objectives (Gao et al., 2021; Xu et al., 2023; Xie et al., 2023; Mao & Nakagawa, 2023; Kim et al., 2023; Zhang et al., 2024a; Li et al., 2025; Vujanic & Rückstieß, 2026) and with generated or targeted data (Lee et al., 2024; Akram et al., 2026). Another family transfers intermediate representations: EMO aligns token relations and hidden states, TALAS combines an endpoint anchor with lower-layer self-distillation, and other methods transfer mixtures of layers or attention (Truong et al., 2025; Dao et al., 2026; Vu et al., 2026; Wang et al., 2024). Such objectives require internal access or choices about which layers to supervise, and cross-model layer correspondence can be fragile (Ko et al., 2023; Yu et al., 2025). GATE-KD instead isolates the terminal-only setting, using cached final teacher embeddings under the TALAS data and evaluation protocol (Section 4.1).

Cross-dimensional representation alignment. Learned projectors provide a common space for teacher and student embeddings, but their parameterization and normalization affect optimization (Zhao et al., 2022; Miles & Mikolajczyk, 2024; Chen et al., 2023). TCS instead obtains fixed lower-dimensional targets using PCA (Zhou et al., 2025). After reduction, orthogonal changes of basis preserve inner products, so a representation belongs to an equivalence class rather than a unique coordinate system (Kornblith et al., 2019; Williams et al., 2021; Maystre et al., 2025). Han (2026) formalizes this principle and argues that absolute coordinates are not teacher content. We begin from the same premise but ask an optimization question left after rank reduction: which student-relative member of the equivalence class should serve as the fixed pointwise target? Bhattacharai et al. (2025) remove orientation by minimizing a Procrustes distance inside the distillation loss. GATE-KD instead selects one corpus-level representative relative to the initial student and exposes it persistently during training; Table 3 compares the two interfaces. Unlike Matryoshka learning and MIPIC, which train nested widths within one model (Kusupati et al., 2022; Huy, 2026), our setting transfers an external teacher whose coordinates and width are unrelated to the student’s.

Coordinate-free structural objectives. Relational, similarity-preserving, and contrastive distillation transfer pairwise distances, Gram matrices, or positive–negative relations (Park et al., 2019; Tung & Mori, 2019; Tian et al., 2020); related work also imposes graph- or geometry-aware constraints (Joshi et al., 2024; Burykina et al., 2025; Zhou et al., 2026). Persistent homology provides multiscale summaries of point-cloud structure and has been used for topological distillation or regularization (Kim et al., 2024; Jeon et al., 2024; Heiter et al., 2023). For sentence embeddings, it has primarily been an analysis tool (Das et al., 2021). GATE-KD uses sorted degree-zero persistence death times only as an optional native-space auxiliary; the gauge-fixed endpoint target provides the primary sample-level supervision.

3 METHOD

Let $f_T : \mathcal{X} \rightarrow \mathbb{R}^{d_T}$ be a frozen teacher and $f_S(\cdot; \theta) : \mathcal{X} \rightarrow \mathbb{R}^{d_S}$ a student with $d_T > d_S$. We run the teacher once over an unlabeled corpus $\{x_i\}_{i=1}^N$ and cache its normalized final embeddings

$$\mathbf{t}_i = \text{norm}(f_T(x_i)), \quad \text{norm}(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|_2}. \quad (1)$$

The corresponding deployed student embedding is $\mathbf{z}_i(\theta) = \text{norm}(f_S(x_i; \theta))$. Both are represented as row vectors.

GATE-KD first selects a fixed student-width subspace with P_{PCA} , then orients it toward the student with a closed-form $R^{(e)}$. Neither matrix receives gradients; only θ is learned. The construction is summarized in Algorithm 1.

3.1 STUDENT-ALIGNED TARGET CONSTRUCTION

Selecting a student-width subspace. Let $\mu = N^{-1} \sum_i \mathbf{t}_i$. We fit the leading d_S principal directions of the centered teacher cache and collect them as $P_{\text{PCA}} \in \mathbb{R}^{d_T \times d_S}$, where $P_{\text{PCA}}^\top P_{\text{PCA}} = I_{d_S}$. We then construct student-width targets

$$\mathbf{y}_i = \text{norm}(\mathbf{t}_i P_{\text{PCA}}) \in \mathbb{S}^{d_S-1}. \quad (2)$$

Centering is used only to fit the directions; we apply neither mean subtraction nor whitening. PCA is a deterministic default, not a requirement: the next step applies to any fixed student-width target matrix (Appendix B.1).

Choosing a student-relative orientation. Dimensional reduction determines which teacher variation is retained but not how the resulting coordinates should be oriented. If $Y \in \mathbb{R}^{N \times d_S}$ collects the rows $\mathbf{y}_i$, then

$$[Y]_{O(d_S)} = \{YR : R \in O(d_S)\} \quad (3)$$

contains equally valid orientations of the reduced targets. Every member preserves the same pairwise geometry because $(YR)(YR)^\top = YY^\top$, yet a pointwise endpoint loss assigns different gradients to different members. GATE-KD therefore selects one orientation relative to the student rather than learning an additional map.

Let $\mathcal{C}$ be a fixed gauge-fit set, equal to the complete distillation corpus in our experiments. At the start of epoch e , we encode $\mathcal{C}$ with the student in evaluation mode and collect the embeddings in $Z_{\mathcal{C}}^{(e)}$. For the matching rows $Y_{\mathcal{C}}$, we solve the orthogonal Procrustes problem (Schönemann, 1966):

$$U\Sigma V^\top = \text{svd}\left(Y_{\mathcal{C}}^\top Z_{\mathcal{C}}^{(e)}\right), \quad (4)$$

$$R^{(e)} = UV^\top \in \arg \min_{R \in O(d_S)} \left\| Y_{\mathcal{C}} R - Z_{\mathcal{C}}^{(e)} \right\|_F^2.$$

The target for sentence x_i during epoch e is

$$\boldsymbol{\tau}_i^{(e)} = \mathbf{y}_i R^{(e)}. \quad (5)$$

All minibatches in epoch e share this target, and $R^{(e)}$ is constant during student updates. It acts only on cached teacher targets and is absent from the deployed student. In particular, $R^{(0)}$ is an interface for one pretrained student checkpoint, not a transferable teacher coordinate frame.

A corpus-global construction is more restrictive than minimizing over orientation independently within each minibatch. For a *fixed* pair of representations (Y, Z) , this distinction in attainable objective values can be stated exactly.

Proposition 1 (Batchwise Procrustes relaxation). *For any partition $\{\mathcal{B}_k\}_{k=1}^K$ of the corpus, let $C_k = Y_{\mathcal{B}_k}^\top Z_{\mathcal{B}_k}$. The locally and globally rotation-minimized mean cosine losses satisfy*

$$\mathcal{L}_{\text{local}} = 1 - \frac{1}{N} \sum_k \|C_k\|_* \leq 1 - \frac{1}{N} \left\| \sum_k C_k \right\|_* = \mathcal{L}_{\text{global}}, \quad (6)$$

where $\|\cdot\|_*$ is the nuclear norm. Equality holds if and only if one orthogonal rotation is simultaneously optimal for every minibatch.

Proof. For any matrix C , orthogonal Procrustes gives $\|C\|_* = \max_{R \in O(d_S)} \text{tr}(R^\top C)$. Maximizing the sum with one R cannot exceed maximizing each term with its own R_k , which yields Equation 6. Equality requires one R to maximize every term, and this condition is sufficient. $\square$

This is a statement about fixed (Y, Z) , not training or generalization. A local zero loss may use different R_k across batches, preserving each within-batch Gram matrix while changing cross-batch similarities through $R_a R_b^\top$. Table 3 changes additional factors and is not a causal test of this freedom.

A learned map can leave the deployed representation non-identifiable. For any invertible A , setting $\mathbf{z}_i = \text{norm}(\mathbf{y}_i A)$ and learning $W = A^{-1}$ gives $\text{norm}(\mathbf{z}_i W) = \mathbf{y}_i$, hence zero mapped cosine loss, while $ZZ^\top$ generally differs from $YY^\top$ (Bhattarai et al., 2025). GATE-KD instead restricts the target-side transformation to $O(d_S)$, which preserves $YY^\top$, and applies the loss directly to the embedding used at inference. This identifies the geometric freedom excluded by our interface, not a claim that projectors are generally ineffective.

3.2 OPTIONAL NATIVE-SPACE H_0 AUXILIARY

The gauge-fixed endpoint target provides the primary sample-level supervision. As an optional coordinate- and dimension-independent signal, we also match a coarse summary of connectivity scales in the original teacher and student spaces. For a minibatch $\mathcal{B}$ of size B , we use chord distance

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{2 - 2\langle \mathbf{u}, \mathbf{v} \rangle}. \quad (7)$$

The $B - 1$ finite death times of degree-zero Vietoris–Rips persistence equal the edge weights of the point cloud’s minimum spanning tree (Hofer et al., 2020). Let $q_{(1)}^S \leq \dots \leq q_{(B-1)}^S$ and $q_{(1)}^T \leq \dots \leq q_{(B-1)}^T$ be the sorted death times computed from the student embeddings $\{\mathbf{z}_i : i \in \mathcal{B}\}$ and the original teacher embeddings $\{\mathbf{t}_i : i \in \mathcal{B}\}$. We define

$$\mathcal{L}_{H_0} = \frac{1}{B-1} \sum_{j=1}^{B-1} \left(q_{(j)}^S - q_{(j)}^T \right)^2. \quad (8)$$

The teacher summary uses $\mathbf{t}_i$, so dimensions and orientation may differ. MST selection and sorting are detached; gradients flow through selected student edges. We use $B = 128$, FP32 distances, $\varepsilon = 10^{-7}$, and SciPy’s upper-triangle input order to break ties. The loss matches only the distribution of connectivity scales; Algorithm 2 gives details.

3.3 STUDENT OPTIMIZATION

For the same minibatch $\mathcal{B}$, the endpoint loss is

$$\mathcal{L}_{\text{end}}^{(e)} = \frac{1}{B} \sum_{i \in \mathcal{B}} \left(1 - \langle \mathbf{z}_i, \boldsymbol{\tau}_i^{(e)} \rangle \right). \quad (9)$$

The complete objective is

$$\mathcal{L}^{(e)} = \mathcal{L}_{\text{end}}^{(e)} + \lambda_{\text{topo}} \mathcal{L}_{H_0}. \quad (10)$$

The headline results use $\lambda_{\text{topo}} = 0.5$; the endpoint-only control sets it to zero. Our main recipe refits the corpus-global orientation after each non-final epoch; the fixed- $R^{(0)}$ schedule is evaluated as a control. A refresh requires one student pass over $\mathcal{C}$ and an SVD of a $d_S \times d_S$ matrix, with no backpropagation through either operation. Teacher embeddings and the PCA subspace are computed once, so ordinary minibatch updates require no online teacher pass. At inference, GATE-KD uses only the student’s normalized final embedding and introduces no additional parameters or computation.

Algorithm 1 GATE-KD training with a non-trainable target-side interface

Require: Frozen teacher f_T ; initialized student $f_S(\cdot; \theta)$; corpus $\mathcal{C}$; epochs E ; refresh epochs $\mathcal{E}_R \subseteq \{0, \dots, E-1\}$ with $0 \in \mathcal{E}_R$; $\lambda_{\text{topo}} \geq 0$.

- 1: Cache $\mathbf{t}_i \leftarrow \text{norm}(f_T(x_i))$ for every $x_i \in \mathcal{C}$.
- 2: Fit P_{PCA} once and construct $\mathbf{y}_i \leftarrow \text{norm}(\mathbf{t}_i P_{\text{PCA}})$.
- 3: **for** $e = 0, \dots, E-1$ **do**
- 4: **if** $e \in \mathcal{E}_R$ **then**
- 5: In evaluation mode and without gradients, encode $\mathcal{C}$ to obtain $Z_{\mathcal{C}}^{(e)}$.
- 6: Compute $USV^\top \leftarrow \text{svd}(Y_{\mathcal{C}}^\top Z_{\mathcal{C}}^{(e)})$ and set $R^{(e)} \leftarrow UV^\top$.
- 7: **else**
- 8: Set $R^{(e)} \leftarrow R^{(e-1)}$.
- 9: **end if**
- 10: **for** each minibatch $\mathcal{B} \subset \mathcal{C}$ **do**
- 11: Compute $\mathbf{z}_i(\theta)$ and $\tau_i^{(e)} \leftarrow \mathbf{y}_i R^{(e)}$ for $i \in \mathcal{B}$.
- 12: Compute $\mathcal{L}_{\text{end}}^{(e)}$ using Equation 9; set $\mathcal{L}_{H_0} \leftarrow 0$ when $\lambda_{\text{topo}} = 0$, and otherwise use Equation 8.
- 13: Update only θ using $\nabla_\theta(\mathcal{L}_{\text{end}}^{(e)} + \lambda_{\text{topo}} \mathcal{L}_{H_0})$.
- 14: **end for**
- 15: **end for**
- 16: **return** $f_S(\cdot; \theta)$.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Teacher–student configurations. We adopt the three teacher–student pairs evaluated by TALAS (Dao et al., 2026). They comprise Qwen3-Embedding-4B $\rightarrow$ BERT-base, BGE-M3 $\rightarrow$ MiniLMv2-L6-H768, and Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-L6-H384 (Devlin et al., 2019; Wang et al., 2021; Chen et al., 2024; Zhang et al., 2025). These settings span students from 22M to 109M parameters and embedding bottlenecks of 2560 $\rightarrow$ 768, 1024 $\rightarrow$ 768, and 1024 $\rightarrow$ 384, respectively. The Qwen teachers use last-token pooling, whereas BGE-M3 uses CLS pooling.

Distillation corpus. Following TALAS (Dao et al., 2026), every method uses the same corpus of about 5K unlabeled training texts from each of Emotion, WiC, and STS-B; no labels are used. We call these three benchmarks source tasks and the other six non-source tasks. “Non-source” denotes only absence from this corpus, not unrestricted domain shift.

Baselines. The baselines span relational transfer (RKD), multi-stage endpoint distillation (Stella), cross-tokenizer or dual-space transfer (CDM, DSKD), and hidden-state alignment (EMO, TALAS) (Park et al., 2019; Zhang et al., 2024a; Chen et al., 2025; Zhang et al., 2024b; Truong et al., 2025; Dao et al., 2026). All share the corpus, model pairs, evaluator, five-epoch budget, batch size 128, and learning-rate grid while retaining method-specific components. This is a controlled-budget comparison, not an upper bound after per-pair tuning; full implementation choices appear in Appendix A.

Evaluation protocol. We evaluate frozen embeddings on classification (Banking77, Emotion, Tweet; macro-F1), pair classification (MRPC, SciTail, WiC; average precision from cosine similarity), and semantic similarity (SICK, STS12, STS-B; Spearman correlation). Metrics are multiplied by 100 and averaged over three seeds. Avg. is the unweighted mean of nine heterogeneous task metrics and is therefore a descriptive index; Src and Non-src separately average the three source and six non-source tasks.

4.2 MAIN RESULTS

GATE-KD has the highest numerical Avg. in all three configurations, exceeding the next-best baseline by 0.38, 0.48, and 1.59 points. The first two are small three-seed differences; the compact result is both larger and broader. Relative to TALAS, GATE-KD is higher on 5/9, 6/9, and 8/9 task means,

Table 1: Nine-task results (mean $\pm$ sample standard deviation, three seeds). Full GATE-KD uses $\lambda_{\text{topo}} = 0.5$; $\dagger$ is the compact endpoint-only aggregate (excluded from task-cell ranking).

Method	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Averages		
	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	Src	Non-src	All
(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)												
Teacher	93.26	71.04	69.08	84.80	91.77	66.79	83.43	82.55	86.87	74.25	84.48	81.07
Student base	84.33	67.08	47.79	77.36	67.74	59.22	42.43	21.54	20.29	42.43	60.08	54.20
RKD	89.76 $\pm$ 0.11	70.90 $\pm$ 0.31	59.98 $\pm$ 0.05	81.94 $\pm$ 0.26	75.59 $\pm$ 0.13	69.02 $\pm$ 0.18	72.93 $\pm$ 0.07	67.84 $\pm$ 0.18	73.72 $\pm$ 0.08	67.57 $\pm$ 0.05	76.49 $\pm$ 0.07	73.52 $\pm$ 0.06
Stella	88.48 $\pm$ 0.30	68.94 $\pm$ 0.17	53.69 $\pm$ 0.50	83.95 $\pm$ 0.10	75.39 $\pm$ 0.54	71.93 $\pm$ 0.09	69.87 $\pm$ 0.64	64.39 $\pm$ 0.21	70.86 $\pm$ 0.36	65.49 $\pm$ 0.17	75.17 $\pm$ 0.13	71.95 $\pm$ 0.06
CDM	88.13 $\pm$ 0.23	69.03 $\pm$ 0.20	52.83 $\pm$ 0.50	83.00 $\pm$ 0.06	77.10 $\pm$ 0.27	71.84 $\pm$ 0.09	67.89 $\pm$ 0.37	61.06 $\pm$ 0.37	69.19 $\pm$ 0.42	64.62 $\pm$ 0.28	74.37 $\pm$ 0.12	71.12 $\pm$ 0.18
DSKD	87.97 $\pm$ 0.16	68.90 $\pm$ 0.28	53.07 $\pm$ 0.31	83.03 $\pm$ 0.15	76.98 $\pm$ 0.59	71.86 $\pm$ 0.08	67.77 $\pm$ 0.66	61.11 $\pm$ 0.26	69.55 $\pm$ 0.12	64.83 $\pm$ 0.12	74.29 $\pm$ 0.14	71.14 $\pm$ 0.10
EMO	86.89 $\pm$ 0.07	68.99 $\pm$ 0.11	52.02 $\pm$ 1.09	79.74 $\pm$ 0.12	76.37 $\pm$ 0.35	69.62 $\pm$ 0.16	67.89 $\pm$ 0.09	56.46 $\pm$ 0.47	64.54 $\pm$ 0.50	62.06 $\pm$ 0.49	72.72 $\pm$ 0.15	69.17 $\pm$ 0.24
TALAS	<u>90.53</u> $\pm$ 0.09	<u>73.77</u> $\pm$ 0.21	<u>68.47</u> $\pm$ 0.34	85.38 $\pm$ 0.05	81.83 $\pm$ 0.21	69.59 $\pm$ 0.14	<u>76.79</u> $\pm$ 0.06	<u>70.79</u> $\pm$ 0.19	78.50 $\pm$ 0.08	72.19 $\pm$ 0.07	<u>79.85</u> $\pm$ 0.07	<u>77.29</u> $\pm$ 0.04
GATE-KD	90.60 $\pm$ 0.21	73.86 $\pm$ 0.10	69.48 $\pm$ 0.22	<u>85.31</u> $\pm$ 0.11	<u>80.23</u> $\pm$ 0.13	68.19 $\pm$ 0.12	77.80 $\pm$ 0.14	75.15 $\pm$ 0.07	<u>78.44</u> $\pm$ 0.16	<u>72.04</u> $\pm$ 0.08	80.49 $\pm$ 0.03	77.67 $\pm$ 0.03
(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)												
Teacher	93.49	73.75	69.03	85.80	91.87	61.46	79.18	78.74	84.86	71.78	83.81	79.80
Student base	81.18	71.30	48.85	79.66	65.06	60.89	49.46	26.76	24.12	44.62	62.24	56.36
RKD	89.14 $\pm$ 0.09	73.29 $\pm$ 0.09	<u>62.23</u> $\pm$ 0.27	84.35 $\pm$ 0.06	78.74 $\pm$ 0.11	65.02 $\pm$ 0.10	73.20 $\pm$ 0.19	65.02 $\pm$ 0.29	71.97 $\pm$ 0.38	66.41 $\pm$ 0.07	77.29 $\pm$ 0.06	73.66 $\pm$ 0.04
Stella	86.82 $\pm$ 0.28	70.06 $\pm$ 0.12	56.61 $\pm$ 0.74	83.02 $\pm$ 0.12	74.63 $\pm$ 0.12	69.72 $\pm$ 0.05	69.08 $\pm$ 0.20	63.14 $\pm$ 0.28	67.10 $\pm$ 0.10	64.47 $\pm$ 0.27	74.46 $\pm$ 0.05	71.13 $\pm$ 0.11
CDM	87.82 $\pm$ 0.19	69.99 $\pm$ 0.09	58.32 $\pm$ 0.51	83.44 $\pm$ 0.07	75.56 $\pm$ 0.40	<u>69.70</u> $\pm$ 0.02	65.93 $\pm$ 0.16	60.84 $\pm$ 0.34	65.31 $\pm$ 0.15	64.45 $\pm$ 0.13	73.93 $\pm$ 0.06	70.77 $\pm$ 0.02
DSKD	87.70 $\pm$ 0.16	70.35 $\pm$ 0.32	59.71 $\pm$ 0.61	82.99 $\pm$ 0.18	75.58 $\pm$ 0.12	69.59 $\pm$ 0.04	65.36 $\pm$ 0.37	61.21 $\pm$ 0.31	65.02 $\pm$ 0.18	64.77 $\pm$ 0.27	73.87 $\pm$ 0.10	70.83 $\pm$ 0.11
EMO	87.54 $\pm$ 0.18	71.03 $\pm$ 0.14	58.52 $\pm$ 0.30	82.46 $\pm$ 0.18	74.73 $\pm$ 0.58	69.20 $\pm$ 0.22	64.71 $\pm$ 0.55	56.89 $\pm$ 0.27	61.25 $\pm$ 0.03	62.99 $\pm$ 0.19	72.89 $\pm$ 0.10	69.59 $\pm$ 0.12
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GATE-KD	90.17 $\pm$ 0.05	<u>73.74</u> $\pm$ 0.24	64.83 $\pm$ 0.72	<u>86.42</u> $\pm$ 0.09	80.97 $\pm$ 0.08	65.47 $\pm$ 0.09	76.77 $\pm$ 0.03	68.27 $\pm$ 0.04	77.63 $\pm$ 0.12	69.31 $\pm$ 0.21	79.39 $\pm$ 0.06	76.03 $\pm$ 0.11
(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)												
Teacher	92.51	71.28	65.80	83.38	90.09	66.84	80.65	77.11	84.59	72.41	82.50	79.14
Student base	67.03	69.45	42.90	78.39	64.55	59.58	47.60	21.96	22.08	41.52	58.16	52.62
RKD	86.76 $\pm$ 0.07	69.87 $\pm$ 0.21	56.31 $\pm$ 0.35	81.70 $\pm$ 0.03	76.03 $\pm$ 0.20	68.56 $\pm$ 0.22	69.21 $\pm$ 0.18	64.26 $\pm$ 0.19	70.25 $\pm$ 0.21	65.04 $\pm$ 0.20	74.64 $\pm$ 0.09	71.44 $\pm$ 0.10
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DSKD	84.76 $\pm$ 0.10	69.56 $\pm$ 0.20	53.14 $\pm$ 0.29	81.15 $\pm$ 0.01	72.81 $\pm$ 0.09	68.33 $\pm$ 0.04	63.78 $\pm$ 0.06	60.24 $\pm$ 0.19	63.53 $\pm$ 0.18	61.67 $\pm$ 0.03	72.05 $\pm$ 0.06	68.59 $\pm$ 0.04
EMO	83.54 $\pm$ 0.31	69.31 $\pm$ 0.06	54.48 $\pm$ 0.27	80.87 $\pm$ 0.13	72.00 $\pm$ 0.16	67.89 $\pm$ 0.09	62.19 $\pm$ 0.16	57.04 $\pm$ 0.17	61.24 $\pm$ 0.23	61.20 $\pm$ 0.09	70.82 $\pm$ 0.07	67.62 $\pm$ 0.02
TALAS	86.39 $\pm$ 0.13	<u>71.81</u> $\pm$ 0.11	<u>58.40</u> $\pm$ 0.30	84.25 $\pm$ 0.03	<u>79.29</u> $\pm$ 0.11	66.82 $\pm$ 0.02	<u>70.42</u> $\pm$ 0.10	<u>68.08</u> $\pm$ 0.22	<u>73.34</u> $\pm$ 0.10	<u>66.19</u> $\pm$ 0.11	<u>76.71</u> $\pm$ 0.04	<u>73.20</u> $\pm$ 0.06
GATE-KD	87.65 $\pm$ 0.06	72.32 $\pm$ 0.24	63.04 $\pm$ 0.28	<u>84.20</u> $\pm$ 0.06	81.60 $\pm$ 0.05	68.07 $\pm$ 0.12	72.48 $\pm$ 0.08	68.96 $\pm$ 0.18	74.76 $\pm$ 0.14	68.62 $\pm$ 0.21	77.87 $\pm$ 0.02	74.79 $\pm$ 0.06
GATE-KD endpoint only †											67.68 $\pm$ 0.19	77.55 $\pm$ 0.07
											74.26 $\pm$ 0.02	

respectively (Appendix D.1), and ranks first in 18 of 27 task–configuration comparisons overall. Pair classification remains mixed.

In configuration (c), endpoint-only GATE-KD reaches 74.26 ± 0.02 , already 1.06 points above TALAS; H_0 adds 0.53 points. Thus the endpoint interface, not the auxiliary, supplies most of the compact gain. The result also shows that intermediate teacher states were not required in this comparison, without implying that intermediate supervision is generally inferior.

On five corpus-disjoint BEIR retrieval benchmarks, GATE-KD ranks first on average in (b) and (c) and second in (a), although all students remain below their teachers (Appendix D.3). This is a transfer guardrail, not a substitute for benchmark-disjoint general-text distillation.

4.3 WHAT MAKES THE INTERFACE WORK?

We use the compact configuration (c) to compare fixed versus learned interfaces, persistent versus stepwise alignment, and fixed versus refreshed targets. Additional controls appear in Appendix B.

4.3.1 DOES THE INTERFACE NEED TO BE LEARNED?

All rows in Table 2 use endpoint-only supervision. The learned controls train one bias-free linear map in either direction under the same AdamW schedule; the fixed controls construct targets before training.

PCA–Procrustes reaches 74.28 Avg., versus 69.33 for the stronger learned map and 69.35 for un-aligned PCA. A fixed target is therefore not enough; its fitted orientation matters. This result is limited to two bias-free linear maps and does not rank stronger learned projector families.

4.3.2 PERSISTENT CORPUS-LEVEL TARGETS VERSUS STEPWISE INVARIANT MATCHING

Table 3 varies the orientation treatment under the full objective. Its invariant row re-solves Procrustes in every minibatch; unlike the fixed rows, it couples local estimation, a moving target, and potentially

Table 2: Compact endpoint-interface ablation (mean $\pm$ sample standard deviation, three seeds).

Target map	Gauge	Retained energy $\uparrow$	Avg. Src $\uparrow$	Avg. Non-src $\uparrow$	Avg. $\uparrow$
Learned $T \rightarrow S$	—	N/A	54.34 $\pm$ 0.68	65.02 $\pm$ 0.25	61.46 $\pm$ 0.39
Learned $S \rightarrow T$	—	N/A	62.54 $\pm$ 0.36	72.72 $\pm$ 0.31	69.33 $\pm$ 0.32
PCA	None	0.928	62.49 $\pm$ 0.19	72.78 $\pm$ 0.07	69.35 $\pm$ 0.04
PCA	Random orthogonal	0.928	63.68 $\pm$ 0.10	73.72 $\pm$ 0.09	70.37 $\pm$ 0.08
Random subspace	None	0.381	64.44 $\pm$ 0.14	74.16 $\pm$ 0.04	70.92 $\pm$ 0.07
PCA	Procrustes	0.928	67.76 $\pm$ 0.05	77.54 $\pm$ 0.07	74.28 $\pm$ 0.03

Table 3: Compact orientation treatments under the full objective (mean $\pm$ sample standard deviation, three seeds).

Orientation treatment	Updates	Config. (c) Avg. $\uparrow$
None (fixed PCA target)	0	72.51 $\pm$ 0.05
Random orthogonal	0	73.02 $\pm$ 0.05
Per-minibatch invariant	Every step	72.74 $\pm$ 0.07
Fixed $R^{(0)}$	0	74.63 $\pm$ 0.07
One refit	1	74.64 $\pm$ 0.06
Epoch-wise Procrustes	4	74.79 $\pm$ 0.06

rank-deficient cross-covariance. It is therefore an interface comparison, not a single-factor test of locality.

Raw PCA, a random fixed orientation, and fitted $R^{(0)}$ score 72.51, 73.02, and 74.63. Fitting the orientation therefore adds 1.61 points over a random fixed gauge and 2.12 over raw coordinates. This supports checkpoint-conditioned presentation, not information in the rotation itself.

The stepwise invariant control scores 72.74 ± 0.07 , 1.89 below fixed $R^{(0)}$. Proposition 1 characterizes its extra batchwise freedom for fixed representations but does not predict this downstream gap; moving targets and minibatch rank remain confounds. A memory-bank or periodic full-corpus invariant control is needed to separate them.

One refit scores 74.64 and four refits 74.79, only 0.01 and 0.16 above $R^{(0)}$. Representative selection matters more than repeated alignment here. Aligned random subspaces also remain within 0.34 points of PCA (Appendix B.1), so the evidence favors a stable, student-aligned target over a particular high-energy subspace.

5 CONCLUSION

GATE-KD exposes a hidden choice in cross-dimensional distillation: projection selects what teacher variation remains, but not how its coordinates are presented to the student. A corpus-level Procrustes fit resolves this ambiguity before optimization, enabling direct supervision of the deployed embedding without a trainable map or layer correspondence.

The compact controls provide the clearest evidence. The fixed endpoint interface exceeds two learned linear maps by at least 4.95 points; one fitted orientation is within 0.16 points of four refits; and aligned random subspaces approach PCA despite much lower retained energy. Across three model pairs, the full recipe has the highest numerical nine-task average, while endpoint supervision supplies most of its gain and H_0 remains auxiliary.

The scope is deliberately narrower than universal embedding transfer. Results use three seeds and task-adapted text; projector controls cover only two linear maps; and the minibatch-invariant comparison also changes target stationarity and fit rank. Proposition 1 explains the static objective relaxation, not the downstream gap. Paired multi-seed analysis, a stronger global-invariant control, general-text distillation, and end-to-end timing remain the highest-priority tests.

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A TRAINING DETAILS

Baseline implementation policy. We use an author-released public implementation whenever one is available. For methods without public code, we reimplement the objective and training procedure described in the corresponding paper. All implementations are integrated into the same training and evaluation framework so that the distillation corpus, teacher–student checkpoints, batch construction, and downstream evaluators are shared. We retain method-specific components, such as Stella’s two-stage schedule and TALAS’s adaptive SAM optimizer, rather than replacing them with GATE-KD’s recipe.

The retained objective settings are as follows. RKD uses distance and angle weights 25 and 50, with its in-batch task term weighted 0.1; this task weight is the one deliberate departure from the published value of 1 under our five-epoch budget. Stella retains its 2 + 3-epoch two-stage schedule and stage weights (0.5, 2.0, 0.5) for cosine, similarity, and triplet terms. CDM uses $\alpha_{\text{DTW}} = 0.5$ and unit CLS weight; DSKD uses $\alpha_{\text{DTW}} = 1$ with bidirectional cross-attention; EMO uses two student layers, unit attention and optimal-transport weights, and $\alpha_{\text{OT}} = 0.1$. TALAS retains its final-two-layer structural term, teacher anchor, adaptive SAM with $\rho = 0.05$, and weights $(w_{\text{task}}, w_{\text{KD}}, w_{\text{struct}}) = (0.001, 0.75, 1)$. GATE-KD uses only its endpoint and H_0 terms in the main comparison. These choices preserve each baseline’s defining mechanism, but the common budget below changes published batch sizes or durations where they differ. We therefore evaluate controlled implementations under one budget, not the best performance attainable after method- and model-pair-specific tuning.

All main-comparison runs use a maximum sequence length of 256 and seeds 42, 43, and 44. We fix the batch size at 128 and the total training duration at five epochs for every method; Stella’s two stages comprise two and three epochs. Learning rate is the only optimization hyperparameter selected separately for each method. Selection is performed on validation data before the three-seed test sweep; the selected rate is then held fixed across all three teacher–student configurations and seeds, and the final-epoch checkpoint is evaluated once on the test splits, so no test score enters learning-rate or checkpoint selection. Every method used the common grid $\{1, 2, 3, 5, 7\} \times 10^{-5}$ and no other per-method tuning. All methods use AdamW with weight decay 0.01 and a cosine schedule ending at 2×10^{-6} . TALAS wraps AdamW in adaptive SAM. For GATE-KD, $\lambda_{\text{topo}} = 0.5$, the H_0 cloud is the current training batch, the gauge is refitted after each non-final epoch, and gauge fitting uses the full distillation corpus. These settings are fixed before test evaluation; the sensitivity sweeps are diagnostic rather than part of model selection. Table 4 lists the selected learning rates, optimizers, and warm-up ratios.

Table 4: Training hyperparameters of the main experiments.

Method	Epochs	Batch	Learning rate	Optimizer	Warm-up
RKD	5	128	7×10^{-5}	AdamW	0.06
Stella	2 + 3	128	5×10^{-5}	AdamW	0.06
CDM	5	128	2×10^{-5}	AdamW	0.06
DSKD	5	128	2×10^{-5}	AdamW	0.06
EMO	5	128	1×10^{-5}	AdamW	0.10
TALAS	5	128	2×10^{-5}	AdamW + adaptive SAM	0.06
GATE-KD	5	128	7×10^{-5}	AdamW	0.06

Optimization-timing scope. CUDA events measure each minibatch update, including the student forward pass, loss computation, backward pass, and optimizer update. Mean update time and throughput exclude the first ten warm-up steps; cumulative optimization time sums all measured updates across five epochs. One-time preparation, epoch-boundary gauge refitting, model and data loading, and final evaluation fall outside this timed region. The archived sweep includes co-located processes, so the reported values characterize the measured optimization environment rather than isolated-process or end-to-end efficiency.

Within this boundary, GATE-KD records the lowest mean update time and cumulative optimization time and the highest throughput in all three configurations. Two GATE-KD-specific costs fall outside it. The PCA fit is one SVD of the 14,760-row cache, and the teacher cache itself is shared with TALAS, RKD, and Stella. Each of the four epoch-boundary refits re-encodes the gauge-fit corpus

Table 5: Optimization-loop cost on NVIDIA H200 GPUs (not end-to-end training time). Mean $\pm$ sample standard deviation over three seeds; best and second-best values are bold and underlined.

Method	Avg. $\uparrow$	ms/step $\downarrow$	samples/s $\uparrow$	Opt. min $\downarrow$	peak GiB $\downarrow$
(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)					
RKD	73.52 $\pm$ 0.06	119.7 $\pm$ 18.3	1081.0 $\pm$ 179.8	1.20 $\pm$ 0.18	1.72 $\pm$ 0.00
Stella	71.95 $\pm$ 0.06	130.7 $\pm$ 8.1	975.9 $\pm$ 58.8	1.27 $\pm$ 0.08	1.78 $\pm$ 0.01
CDM	71.12 $\pm$ 0.18	340.4 $\pm$ 13.6	374.2 $\pm$ 15.3	3.38 $\pm$ 0.12	9.31 $\pm$ 0.01
DSKD	71.14 $\pm$ 0.10	290.9 $\pm$ 9.0	437.7 $\pm$ 13.3	2.88 $\pm$ 0.09	9.45 $\pm$ 0.00
EMO	69.17 $\pm$ 0.24	721.1 $\pm$ 12.9	176.5 $\pm$ 3.1	7.02 $\pm$ 0.12	9.30 $\pm$ 0.00
TALAS	77.29 $\pm$ 0.04	329.0 $\pm$ 12.5	387.1 $\pm$ 15.0	3.23 $\pm$ 0.11	1.43 $\pm$ 0.00
GATE-KD	77.67 $\pm$ 0.03	83.4 $\pm$ 3.8	1527.2 $\pm$ 70.8	0.84 $\pm$ 0.04	<u>1.72</u> $\pm$ 0.00
(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)					
RKD	73.66 $\pm$ 0.04	77.5 $\pm$ 8.8	1656.4 $\pm$ 201.2	0.79 $\pm$ 0.09	1.07 $\pm$ 0.00
Stella	71.13 $\pm$ 0.11	152.8 $\pm$ 30.8	859.1 $\pm$ 195.6	1.49 $\pm$ 0.30	1.09 $\pm$ 0.00
CDM	70.77 $\pm$ 0.02	127.5 $\pm$ 13.8	1006.4 $\pm$ 115.8	1.30 $\pm$ 0.12	2.15 $\pm$ 0.01
DSKD	70.83 $\pm$ 0.11	115.0 $\pm$ 21.9	1137.2 $\pm$ 241.0	1.14 $\pm$ 0.20	2.22 $\pm$ 0.01
EMO	69.59 $\pm$ 0.12	394.6 $\pm$ 33.4	324.0 $\pm$ 28.3	3.87 $\pm$ 0.35	2.15 $\pm$ 0.01
TALAS	75.55 $\pm$ 0.06	184.4 $\pm$ 4.6	690.2 $\pm$ 17.3	1.82 $\pm$ 0.03	0.85 $\pm$ 0.00
GATE-KD	76.03 $\pm$ 0.11	56.5 $\pm$ 7.7	2281.6 $\pm$ 310.0	0.57 $\pm$ 0.07	<u>1.07</u> $\pm$ 0.00
(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)					
RKD	71.44 $\pm$ 0.10	64.8 $\pm$ 13.7	2012.3 $\pm$ 488.6	0.68 $\pm$ 0.14	0.41 $\pm$ 0.00
Stella	68.57 $\pm$ 0.10	150.7 $\pm$ 15.3	850.0 $\pm$ 81.5	1.47 $\pm$ 0.13	0.42 $\pm$ 0.00
CDM	68.69 $\pm$ 0.27	161.8 $\pm$ 21.6	796.3 $\pm$ 113.1	1.65 $\pm$ 0.20	1.53 $\pm$ 0.00
DSKD	68.59 $\pm$ 0.04	175.4 $\pm$ 8.2	726.4 $\pm$ 33.5	1.78 $\pm$ 0.06	1.56 $\pm$ 0.00
EMO	67.62 $\pm$ 0.02	499.3 $\pm$ 1.4	254.8 $\pm$ 0.7	4.90 $\pm$ 0.01	1.53 $\pm$ 0.00
TALAS	73.20 $\pm$ 0.06	169.8 $\pm$ 15.4	753.7 $\pm$ 71.4	1.69 $\pm$ 0.14	0.34 $\pm$ 0.00
GATE-KD	74.79 $\pm$ 0.06	62.9 $\pm$ 3.1	2025.2 $\pm$ 98.3	0.64 $\pm$ 0.04	<u>0.41</u> $\pm$ 0.00

with the student; charging those forward passes at the training-step throughput of Table 5, an upper bound because it does not subtract the backward pass, adds at most 0.64, 0.43, and 0.49 minutes in configurations (a), (b), and (c). We report this accounting to bound a method-specific omission, not as a directly measured end-to-end time. Teacher caching, PCA, refit I/O, model loading, evaluation, and baseline-specific preprocessing are not jointly timed; consequently we make no claim that Table 5 establishes lower end-to-end training cost.

B ENDPOINT-INTERFACE ANALYSIS

B.1 ROBUSTNESS TO FIXED-SUBSPACE CONSTRUCTION

We isolate teacher projection in configuration (c), holding the full GATE-KD objective (native-space H_0 , $\lambda_{\text{topo}} = 0.5$), student, optimizer, schedule, and epoch-wise Procrustes fixed. These matched reruns need not equal the main comparison numerically. The reference fits PCA directions on the centered cache but applies them without subtracting the mean; mean-subtracted PCA applies this affine centering, and whitening also rescales by inverse square root eigenvalues. Random controls use three independent Haar subspaces of the same rank. Each row averages seeds 42–44, with random draws kept separate.

Table 6: PCA preprocessing and subspace controls in configuration (c). Mean $\pm$ sample standard deviation over three seeds; best values are bold.

Interface	Retained energy $\uparrow$	Avg. Src $\uparrow$	Avg. Non-src $\uparrow$	Avg. $\uparrow$
PCA + Procrustes	0.928	68.62 $\pm$ 0.21	77.87 $\pm$ 0.02	74.79 $\pm$ 0.06
Mean-subtracted PCA	0.928	66.49 $\pm$ 0.12	74.89 $\pm$ 0.02	72.09 $\pm$ 0.05
PCA whitening	0.928	66.44 $\pm$ 0.14	74.16 $\pm$ 0.04	71.59 $\pm$ 0.06
Random subspace (draw 0)	0.381	68.05 $\pm$ 0.16	77.65 $\pm$ 0.08	74.45 $\pm$ 0.11
Random subspace (draw 1)	0.376	68.35 $\pm$ 0.28	77.82 $\pm$ 0.04	74.66 $\pm$ 0.07
Random subspace (draw 2)	0.374	68.58 $\pm$ 0.16	77.40 $\pm$ 0.05	74.46 $\pm$ 0.06
Uncentered SVD	0.945	68.67 $\pm$ 0.08	77.82 $\pm$ 0.03	74.60 $\pm$ 0.03

Mean subtraction lowers Avg. from 74.79 to 72.09 (−2.70), and whitening to 71.59 (−3.20), with both source and non-source averages declining. Aligned random subspaces reach 74.45–74.66, only 0.13–0.34 below PCA despite retaining 0.374–0.381 rather than 0.928 of the teacher energy. The updated uncentered-SVD run reaches 74.60, 0.19 points below PCA overall. Preserving the mean component at application time therefore matters, whereas once the gauge is aligned neither retained energy nor the exact fitted subspace is decisive; PCA is numerically best, but not established as uniquely optimal.

Retained energy and the endpoint objective measure different quantities. The former is raw coordinate reconstruction before row normalization, whereas the latter acts on normalized directions after projection and alignment. A random subspace can therefore discard substantial total variance without necessarily discarding angular distinctions useful to the student. Concentration in a high-dimensional projection and redundancy in the teacher representation are plausible explanations, but we did not measure normalized Gram distortion for the random projections. The result consequently narrows our claim: PCA is a stable deterministic default, while the evidence attributes more importance to student-conditioned orientation than to maximum retained energy.

B.2 ENDPOINT-INTERFACE RANK AT FIXED STUDENT WIDTH

We isolate the subspace selected by the endpoint interface in configuration (c). The student architecture, 384-dimensional output, distillation corpus, optimizer, training schedule, and Procrustes updates remain fixed, while the retained PCA rank varies over $k \in \{64, 128, 256, 384\}$. This experiment uses the endpoint-only objective. Targets with $k < 384$ are zero-padded to the student width before Procrustes alignment, which preserves their Gram matrix and active rank.

Retained energy is measured on the complete teacher cache. On a fixed sample of 2,048 normalized teacher embeddings T , we measure the normalized squared Gram distortion $\|TT^\top - Y_k Y_k^\top\|_F^2 / \|TT^\top\|_F^2$, where Y_k is the normalized rank- k PCA target. We also report the normalized Eckart–Young rank floor $\sum_{j>k} \sigma_j(T)^4 / \sum_j \sigma_j(T)^4$ (Eckart & Young, 1936). The Gram quantities are independent of the orthogonal gauge and are therefore computed once, while downstream scores are averaged over three seeds.

Figure 1 shows monotonic changes in all three quantities. Increasing k from 64 to 384 raises retained teacher energy from 59.32% to 92.84%, reduces normalized Gram distortion from 6.797% to 0.086%, and improves Avg. from 72.57 to 74.27. The intermediate ranks 128 and 256 attain 73.77 and 74.21 Avg., respectively. Most of the downstream benefit is therefore recovered by rank 256: it captures 96% of the 64-to-384 improvement, with the final increase to rank 384 contributing only 0.06 points in this configuration.

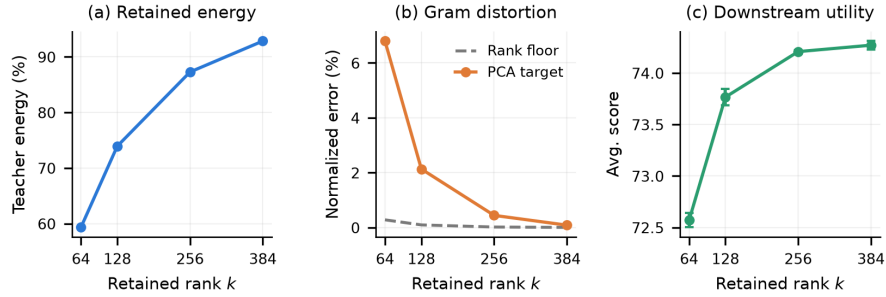


Figure 1: Sensitivity to retained PCA rank at fixed student width in configuration (c), endpoint-only objective, one corpus, four ranks. Targets with $k < 384$ are zero-padded before Procrustes alignment. Downstream scores show mean $\pm$ sample standard deviation over three seeds; retained energy and Gram distortion are gauge-independent and computed once.

PCA distortion remains roughly 21–25 times above the rank floor because its objective differs from normalized Gram approximation: PCA minimizes reconstruction error of the raw coordinates, whereas the Gram quantity compares inner products after re-normalizing each projected row, and

re-normalization amplifies the truncated tail for rows with small retained norm. The floor therefore bounds irreducible error rather than predicting this interface’s distortion, and retained energy should not be read as a proxy for how well the teacher’s angular geometry is preserved; only the Gram distortion column measures that directly. Two further scope notes apply. First, because the padded target has zero coordinates outside its rotated k -dimensional subspace, the cosine endpoint loss drives the student’s remaining $384 - k$ output coordinates toward zero rather than leaving them free, so for $k < 384$ the student effectively operates at rank k despite its 384-dimensional output. Second, the sweep is endpoint-only, in configuration (c), on one corpus; we have not repeated it with the H_0 term or in configurations (a) and (b).

B.3 GAUGE-FIT-SET SIZE

Table 7 varies only the number of corpus examples used to estimate the orthogonal gauge in configuration (c). Reducing the fit set from all 14,760 sentences to 2,048 changes Avg. by 0.28 points. This suggests that the fitted coordinate relation can be estimated from a modest sample under the evaluated protocol; it does not establish out-of-distribution generalization of the gauge.

Table 7: Gauge-fit-set-size sensitivity in configuration (c), reported as mean $\pm$ sample standard deviation over three seeds. Best values are bold.

Fit-set size	Avg. Src $\uparrow$	Avg. Non-src $\uparrow$	Avg. $\uparrow$
2,048	68.36 $\pm$ 0.11	77.59 $\pm$ 0.04	74.51 $\pm$ 0.06
4,096	68.52 $\pm$ 0.08	77.74 $\pm$ 0.05	74.67 $\pm$ 0.02
8,192	68.26 $\pm$ 0.08	77.78 $\pm$ 0.03	74.61 $\pm$ 0.02
14,760	68.62 $\pm$ 0.21	77.87 $\pm$ 0.02	74.79 $\pm$ 0.06

C STRUCTURAL-REGULARIZATION ANALYSIS

C.1 ENDPOINT AND TOPOLOGY SIGNAL DECOMPOSITION

Table 8 separates the endpoint and topology signals. The endpoint-only objective supplies most of the downstream gain, whereas H_0 -only training reduces the connectivity residual without reaching the endpoint-supervised score. Relative to the no-teacher control, endpoint-only training gains 21.66 points, compared with 11.62 points for H_0 alone. Adding native-space H_0 to the endpoint objective yields a further 0.53 points and reduces the native-space H_0 residual from 0.2122 to 0.1281. The H_0 -only row uses $\lambda_{\text{topo}} = 1$, whereas the combined row repeats the $\lambda_{\text{topo}} = 0.5$ main reference; their difference should therefore be read as a signal decomposition rather than a matched weight ablation.

Table 8: Endpoint and topology signal decomposition in configuration (c). Mean $\pm$ sample standard deviation over three seeds; best values are bold.

Objective	λ_{topo}	H_0 residual $\downarrow$	Avg. Src $\uparrow$	Avg. Non-src $\uparrow$	Avg. $\uparrow$
No-teacher control	—	0.3595 $\pm$ 0.0000	41.61 $\pm$ 0.00	58.10 $\pm$ 0.00	52.60 $\pm$ 0.00
H_0 only	1.0	0.1240 $\pm$ 0.0012	58.00 $\pm$ 1.01	67.33 $\pm$ 0.62	64.22 $\pm$ 0.73
Endpoint only	0	0.2122 $\pm$ 0.0004	67.68 $\pm$ 0.19	77.55 $\pm$ 0.07	74.26 $\pm$ 0.02
Endpoint + native-space H_0	0.5	0.1281 $\pm$ 0.0003	68.62 $\pm$ 0.21	77.87 $\pm$ 0.02	74.79 $\pm$ 0.06

C.2 STRUCTURAL-CONTROL DEFINITIONS

Table 9 compares the optional native-space H_0 term with structural alternatives under the same endpoint interface and optimization protocol. The “None” row repeats the endpoint-only reference; the H_0 row repeats configuration (c) from the main comparison.

Table 9: Matched structural controls in configuration (c). Mean $\pm$ sample standard deviation over three seeds; best value is bold.

Auxiliary term	Avg. $\uparrow$	Δ endpoint
None	74.26 $\pm$ 0.02	—
Sorted spanning path	73.66 $\pm$ 0.04	−0.60
kNN distribution	74.47 $\pm$ 0.01	+0.21
Teacher-MST edges	74.33 $\pm$ 0.08	+0.07
Native Gram matching	73.57 $\pm$ 0.01	−0.69
H_0	74.79$\pm$ 0.06	+0.53

The sorted spanning-path control constructs a geometry-independent path through the shuffled batch and matches the independently sorted student and teacher edge lengths. The kNN control similarly matches independently sorted directed nearest-neighbor distances with $k = 1$. Teacher-MST first selects the teacher’s $B - 1$ minimum-spanning-tree edges and then compares student distances on those same sample pairs, retaining edge correspondence. Native Gram matching compares all pairwise cosine similarities. Finally, H_0 matches independently sorted student and teacher MST edge weights and discards edge correspondence. The path, Teacher-MST, and H_0 controls expose $B - 1$ scalar constraints per batch, kNN exposes B , and Gram matching is deliberately denser.

Among the evaluated controls and weights, H_0 produces the largest positive change. The comparison rules out several simple alternatives but does not isolate MST sorting or persistence-specific information: the objectives also differ in graph construction, correspondence, density, and gradient behavior.

C.3 IMPLEMENTATION OF THE H_0 LOSS

Algorithm 2 makes the nonsmooth implementation explicit. The teacher branch is evaluated without gradients. On the student branch, the MST edge selection and sorting permutation are detached, while the selected edge weights remain differentiable. Exact distance ties are passed to SciPy in row-major upper-triangular order. Although the selected active tree can depend on this library tie-break, every valid MST has the same sorted multiset of edge weights; the backward pass follows the particular active tree returned in that forward pass. All reported runs use chord distance, FP32 Gram computation, $\varepsilon = 10^{-7}$ inside the square root, one 128-example cloud per full training batch (and the complete smaller tail batch), and $\lambda_{\text{topo}} = 0.5$ for the combined objective.

Algorithm 2 Native-space H_0 connectivity-scale loss for one minibatch

Require: Row-normalized teacher and student batches T_B and Z_B ; numerical floor ε .

```

1: for  $A \in \{T_B, Z_B\}$  do
2:   Compute  $G \leftarrow AA^\top$  in FP32 and symmetrize  $G \leftarrow (G + G^\top)/2$ .
3:   Compute  $D_{ij} \leftarrow \sqrt{\max(2 - 2G_{ij}, \varepsilon)}$  and set  $D_{ii} \leftarrow 0$ .
4:   On a detached CPU float64 copy, add the same positive constant to every off-diagonal edge
   and select an undirected MST with SciPy from the upper triangle.
5:   Gather the selected weights from the unshifted  $D$  and sort them to obtain  $q_{(1:B-1)}^A$ .
6: end for
7: return  $(B - 1)^{-1} \sum_{j=1}^{B-1} (q_{(j)}^Z - q_{(j)}^T)^2$ .
```

C.4 TOPOLOGY-WEIGHT SENSITIVITY

Table 10 varies the topology weight in configuration (c), with all other settings fixed to the main recipe. Every tested nonzero weight improves over the endpoint-only setting, while values from 0.5 to 1.0 differ by at most 0.24 points.

Table 10: Topology-weight sensitivity in configuration (c), reported as mean $\pm$ sample standard deviation over three seeds. Best values are bold.

λ_{topo}	Avg. Src $\uparrow$	Avg. Non-src $\uparrow$	Avg. $\uparrow$
0	67.54 $\pm$ 0.18	77.54 $\pm$ 0.07	74.21 $\pm$ 0.03
0.5	68.62 $\pm$ 0.21	77.87 $\pm$ 0.02	74.79 $\pm$ 0.06
0.75	68.47 $\pm$ 0.13	77.84 $\pm$ 0.04	74.72 $\pm$ 0.04
1	68.14 $\pm$ 0.06	77.75 $\pm$ 0.03	74.55 $\pm$ 0.03

D ADDITIONAL EVALUATION

D.1 TASK-LEVEL MEAN DIFFERENCES FROM TALAS

Table 11 expands the aggregate comparison into differences of the reported task means. Configuration (a) is mixed: GATE-KD is higher on five tasks, and the overall gain is influenced strongly by STS12. Configuration (b) is higher on six tasks. The compact configuration (c) is the most consistent, with positive differences on eight of nine tasks and the only negative difference being -0.05 on MRPC. These are descriptive differences of three-seed means, not paired confidence intervals or per-task significance tests.

Table 11: Task-level differences of reported means, GATE-KD minus TALAS. Positive values favor GATE-KD. The final column counts positive task-level differences. These are descriptive differences of means over three seeds, not seed-paired uncertainty estimates.

Config.	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	Avg.	Positive
(a)	+0.07	+0.09	+1.01	-0.07	-1.60	-1.40	+1.01	+4.36	-0.06	+0.38	5/9
(b)	+0.34	-0.96	+3.18	-0.20	+0.12	-1.25	+1.37	+1.19	+0.55	+0.48	6/9
(c)	+1.26	+0.51	+4.64	-0.05	+2.31	+1.25	+2.06	+0.88	+1.42	+1.59	8/9

D.2 BROADER 100K TASK-ADAPTED CORPUS

We repeat the complete method sweep on approximately 100K unlabeled texts drawn from the training splits of all nine sentence-level benchmarks. No labels or task annotations are used. Relative to the main comparison, this setting changes both corpus size and dataset coverage. It therefore evaluates robustness in a broader task-adapted setting rather than isolating the effect of corpus size alone.

Table 12: Results with the broader 100K distillation corpus. Distilled-student scores are mean $\pm$ sample standard deviation over three seeds; best and second-best results are bold and underlined.

Method	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Avg.
	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	
(a) Qwen3-Embedding-4B → BERT-base (109M)										
Teacher	93.26	71.04	69.08	84.80	91.77	66.79	83.43	82.55	86.87	81.07
Student base	84.33	67.08	47.79	77.36	67.74	59.22	42.43	21.54	20.29	54.20
RKD	92.23±0.15	71.96±0.18	63.79±0.40	85.00±0.04	80.27±0.38	66.31±0.10	79.32±0.02	71.72±0.16	77.70±0.20	76.48±0.04
Stella	90.02±0.15	71.90±0.22	60.14±0.45	85.96±0.05	78.82±0.27	69.93±0.13	76.53±0.23	69.79±0.12	77.22±0.16	75.59±0.10
CDM	89.16±0.04	69.63±0.23	53.74±0.13	85.57±0.08	71.22±0.19	70.04±0.05	71.26±0.29	66.97±0.21	73.69±0.38	72.36±0.13
DSKD	88.87±0.07	69.05±0.09	54.02±0.56	85.90±0.05	70.02±0.31	69.63±0.21	71.07±0.10	66.52±0.17	73.52±0.01	72.07±0.03
EMO	84.96±0.84	68.93±0.65	54.80±1.52	85.05±0.04	65.75±0.22	68.80±0.13	62.00±2.71	61.75±1.85	69.54±0.37	69.07±0.89
TALAS	93.12±0.13	74.67±0.38	71.42±0.26	86.30±0.01	87.06±0.12	68.09±0.02	80.29±0.03	76.34±0.01	82.26±0.05	79.95±0.08
GATE-KD	93.47±0.12	74.18±0.25	72.50±0.47	85.45±0.08	87.24±0.12	66.57±0.08	81.85±0.06	76.40±0.07	82.52±0.08	80.02±0.03
(b) BGE-M3 → MiniLMv2-H768 (66M)										
Teacher	93.49	73.75	69.03	85.80	91.87	61.46	79.18	78.74	84.86	79.80
Student base	81.18	71.30	48.85	79.66	65.06	60.89	49.46	26.76	24.12	56.36
RKD	91.68±0.06	74.57±0.40	62.45±0.75	85.81±0.08	79.81±0.17	63.57±0.13	76.91±0.05	69.54±0.11	75.56±0.09	75.54±0.10
Stella	88.72±0.41	73.43±0.23	61.63±0.29	86.39±0.08	74.13±0.31	66.98±0.27	73.32±0.07	68.30±0.12	72.66±0.14	73.95±0.04
CDM	87.64±0.23	70.82±0.14	56.37±0.57	85.67±0.08	67.69±0.23	66.89±0.20	70.15±0.48	66.68±0.26	69.49±0.15	71.27±0.10
DSKD	86.69±0.38	69.76±0.22	56.98±0.39	84.92±0.07	66.75±0.17	67.23±0.10	68.26±0.45	66.28±0.02	68.31±0.35	70.58±0.10
EMO	87.17±0.59	71.07±0.22	59.32±0.02	84.25±0.41	67.89±0.92	66.09±0.18	69.04±0.38	62.26±1.77	66.47±0.40	70.40±0.35
TALAS	92.44±0.10	75.58±0.15	65.13±0.21	87.09±0.09	85.39±0.14	64.39±0.12	78.14±0.06	72.56±0.13	81.03±0.05	77.97±0.03
GATE-KD	92.56±0.04	76.30±0.31	65.62±0.29	86.58±0.05	87.20±0.06	63.43±0.02	79.23±0.03	72.69±0.05	80.60±0.13	78.24±0.05
(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)										
Teacher	92.51	71.28	65.80	83.38	90.09	66.84	80.65	77.11	84.59	79.14
Student base	67.03	69.45	42.90	78.39	64.55	59.58	47.60	21.96	22.08	52.62
RKD	89.74±0.04	72.34±0.22	57.99±0.52	84.13±0.12	75.20±0.14	67.49±0.06	74.48±0.03	68.12±0.15	72.55±0.14	73.56±0.09
Stella	85.74±0.22	70.25±0.07	53.48±0.28	84.43±0.04	71.23±0.05	66.51±0.14	70.96±0.28	63.02±0.04	67.42±0.24	70.34±0.02
CDM	85.18±0.08	69.14±0.07	52.30±0.26	84.02±0.04	68.24±0.23	66.57±0.29	69.15±0.13	63.05±0.06	66.55±0.10	69.36±0.08
DSKD	85.04±0.14	69.21±0.13	52.85±0.36	84.34±0.10	67.33±0.24	65.98±0.12	69.01±0.13	63.46±0.21	66.42±0.09	69.29±0.09
EMO	85.08±0.17	68.11±0.10	53.96±0.60	84.38±0.16	65.48±0.16	65.23±0.17	68.61±0.07	60.00±0.82	65.38±0.38	68.47±0.15
TALAS	89.79±0.03	73.87±0.17	61.75±0.61	85.99±0.03	82.07±0.09	65.79±0.04	75.67±0.05	73.91±0.03	78.31±0.04	76.35±0.08
GATE-KD	91.68±0.09	74.43±0.21	66.88±0.36	85.00±0.06	83.46±0.02	67.44±0.04	77.94±0.06	72.17±0.02	78.36±0.07	77.48±0.04

Table 12 shows that GATE-KD attains the highest mean Avg. in all three teacher–student configurations. It exceeds TALAS by 0.07, 0.27, and 1.13 points in configurations (a), (b), and (c), respectively. The margin is narrow for the two larger students and remains largest for the compact student.

D.3 ZERO-SHOT RETRIEVAL

We evaluate the models distilled in the main comparison on ArguAna, SciFact, SciDocs, NFCorpus, and FiQA-2018 using exhaustive cosine ranking. Evaluation uses no instruction prefix or task-specific retrieval training and reports nDCG@10. Retrieval scores are not included in the sentence-level averages of Table 1. Teacher and student-base scores are deterministic, while distilled-student scores are mean $\pm$ sample standard deviation over three seeds. None of these five retrieval corpora contributes text to the distillation set, making this evaluation corpus-disjoint from distillation; however, the models are still trained on the task-adapted sentence corpus, so this is a transfer guardrail rather than a benchmark-disjoint training protocol.

Table 13: Zero-shot retrieval on five BEIR benchmarks (nDCG@10). Distilled-student scores are mean $\pm$ sample standard deviation over three seeds; the best and second-best distilled-student results are bold and underlined.

Method	ArguAna	SciFact	SciDocs	NFCorpus	FiQA-2018	Avg.
<i>(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)</i>						
Teacher	74.66	73.75	25.69	36.23	50.15	52.10
Student base	12.65	0.89	0.38	1.33	0.05	3.06
RKD	39.68 $\pm$ 0.25	9.74 $\pm$ 0.64	1.19 $\pm$ 0.04	4.30 $\pm$ 0.30	2.82 $\pm$ 0.09	11.55 $\pm$ 0.18
Stella	39.12 $\pm$ 0.17	14.29 $\pm$ 0.17	<u>3.79</u> $\pm$ 0.26	6.21 $\pm$ 0.14	5.57 $\pm$ 0.04	13.79 $\pm$ 0.03
CDM	33.62 $\pm$ 0.27	15.11 $\pm$ 0.20	3.28 $\pm$ 0.14	<u>6.26</u> $\pm$ 0.09	5.08 $\pm$ 0.22	12.67 $\pm$ 0.04
DSKD	33.70 $\pm$ 0.39	15.75 $\pm$ 0.49	3.35 $\pm$ 0.11	6.21 $\pm$ 0.17	5.34 $\pm$ 0.24	12.87 $\pm$ 0.27
EMO	21.90 $\pm$ 0.34	8.84 $\pm$ 0.28	1.87 $\pm$ 0.11	4.87 $\pm$ 0.16	2.04 $\pm$ 0.06	7.91 $\pm$ 0.07
TALAS	48.95 $\pm$ 0.17	19.94 $\pm$ 0.09	5.70 $\pm$ 0.12	7.94 $\pm$ 0.08	8.15 $\pm$ 0.02	18.14 $\pm$ 0.07
GATE-KD	<u>45.20</u> $\pm$ 0.32	<u>19.25</u> $\pm$ 0.66	2.81 $\pm$ 0.09	<u>6.79</u> $\pm$ 0.22	<u>6.56</u> $\pm$ 0.21	<u>16.12</u> $\pm$ 0.26
<i>(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)</i>						
Teacher	54.01	64.16	16.37	31.80	41.12	41.49
Student base	9.32	1.15	0.44	1.22	0.08	2.44
RKD	21.27 $\pm$ 0.23	7.84 $\pm$ 0.12	<u>1.96</u> $\pm$ 0.05	4.31 $\pm$ 0.14	2.26 $\pm$ 0.03	7.53 $\pm$ 0.09
Stella	18.46 $\pm$ 0.37	3.11 $\pm$ 0.27	1.25 $\pm$ 0.06	2.29 $\pm$ 0.15	1.66 $\pm$ 0.08	5.35 $\pm$ 0.16
CDM	15.83 $\pm$ 0.19	3.77 $\pm$ 0.38	0.88 $\pm$ 0.02	2.88 $\pm$ 0.16	1.41 $\pm$ 0.21	4.95 $\pm$ 0.15
DSKD	16.25 $\pm$ 0.15	4.05 $\pm$ 0.10	0.88 $\pm$ 0.03	2.96 $\pm$ 0.11	1.59 $\pm$ 0.07	5.15 $\pm$ 0.07
EMO	13.10 $\pm$ 0.27	2.83 $\pm$ 0.17	0.72 $\pm$ 0.04	2.25 $\pm$ 0.07	0.85 $\pm$ 0.09	3.95 $\pm$ 0.08
TALAS	30.14 $\pm$ 0.32	6.55 $\pm$ 0.34	1.90 $\pm$ 0.08	2.82 $\pm$ 0.18	<u>2.83</u> $\pm$ 0.09	8.85 $\pm$ 0.17
GATE-KD	<u>25.30</u> $\pm$ 0.02	11.22 $\pm$ 0.27	3.31 $\pm$ 0.02	4.93 $\pm$ 0.06	4.83 $\pm$ 0.01	9.92 $\pm$ 0.05
<i>(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)</i>						
Teacher	68.11	68.26	15.72	29.73	38.68	44.10
Student base	9.85	0.66	0.32	1.28	0.05	2.43
RKD	30.12 $\pm$ 0.79	<u>5.57</u> $\pm$ 0.76	<u>1.77</u> $\pm$ 0.12	<u>3.34</u> $\pm$ 0.07	<u>1.88</u> $\pm$ 0.06	8.54 $\pm$ 0.33
Stella	24.49 $\pm$ 0.45	2.65 $\pm$ 0.09	1.09 $\pm$ 0.07	2.62 $\pm$ 0.07	1.01 $\pm$ 0.11	6.37 $\pm$ 0.09
CDM	21.09 $\pm$ 0.30	1.66 $\pm$ 0.17	1.13 $\pm$ 0.03	2.57 $\pm$ 0.05	0.89 $\pm$ 0.02	5.47 $\pm$ 0.06
DSKD	21.56 $\pm$ 0.38	1.77 $\pm$ 0.20	1.18 $\pm$ 0.06	2.63 $\pm$ 0.03	0.89 $\pm$ 0.07	5.61 $\pm$ 0.13
EMO	17.18 $\pm$ 0.14	1.41 $\pm$ 0.09	0.92 $\pm$ 0.02	1.96 $\pm$ 0.06	0.70 $\pm$ 0.03	4.44 $\pm$ 0.05
TALAS	<u>35.21</u> $\pm$ 0.52	4.17 $\pm$ 0.09	1.44 $\pm$ 0.04	3.12 $\pm$ 0.10	1.87 $\pm$ 0.07	<u>9.16</u> $\pm$ 0.14
GATE-KD	36.18 $\pm$ 0.09	10.26 $\pm$ 0.21	3.44 $\pm$ 0.06	4.30 $\pm$ 0.06	3.57 $\pm$ 0.03	11.55 $\pm$ 0.06

GATE-KD achieves the highest retrieval average in configurations (b) and (c). It exceeds TALAS by 1.07 and 2.39 points, respectively, and leads on nine of the ten individual retrieval comparisons across those two configurations. In configuration (a), it ranks second to TALAS, with 16.12 versus 18.14 Avg. All distilled students remain substantially below their teachers, indicating that the sentence-level distillation objectives do not fully preserve the teachers’ zero-shot retrieval capability.

E EXPLORATORY REPRESENTATION DIAGNOSTICS

We examine whether final-embedding supervision changes the internal representations of the student even though GATE-KD applies no intermediate-layer objective. The probe contains 512 unique sentences sampled from the validation splits after excluding sentences found in the distillation corpus. For every teacher and student Transformer layer, we compute centered linear CKA between their pooled representations (Kornblith et al., 2019). Results for the trained students are averaged over seeds 42, 43, and 44. We omit the embedding-output layer because special-token pooling makes this representation constant across examples, for which centered CKA is undefined.

Figure 2 shows that the pretrained students already share substantial representation structure with their teachers. Raw CKA alone therefore combines correspondence inherited from pretraining with changes induced by distillation.

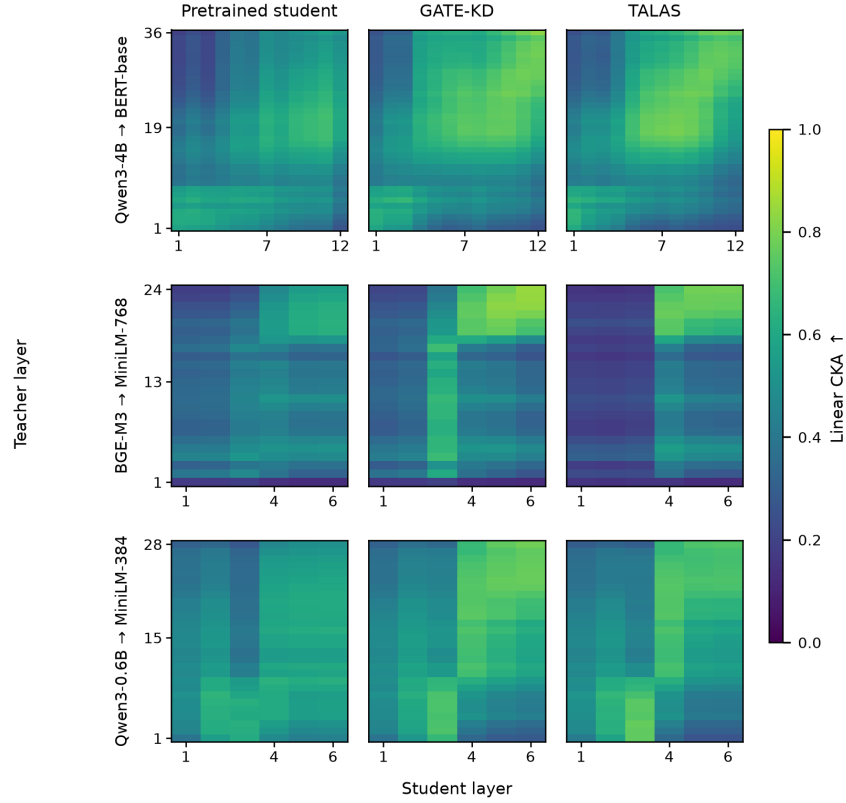


Figure 2: Layerwise linear CKA for the pretrained student, GATE-KD, and TALAS across three configurations. Trained results are averaged over three seeds.

To contextualize representation similarity relative to initialization, Figure 3 subtracts the pretrained-student CKA matrix from the mean matrix after distillation. Final-to-final CKA is higher for GATE-KD than for TALAS in every configuration: 0.8266, 0.8447, and 0.7845 versus 0.7741, 0.7901, and 0.7333 in configurations (a), (b), and (c), a difference of 0.051 to 0.055 that is also present in the change from initialization.

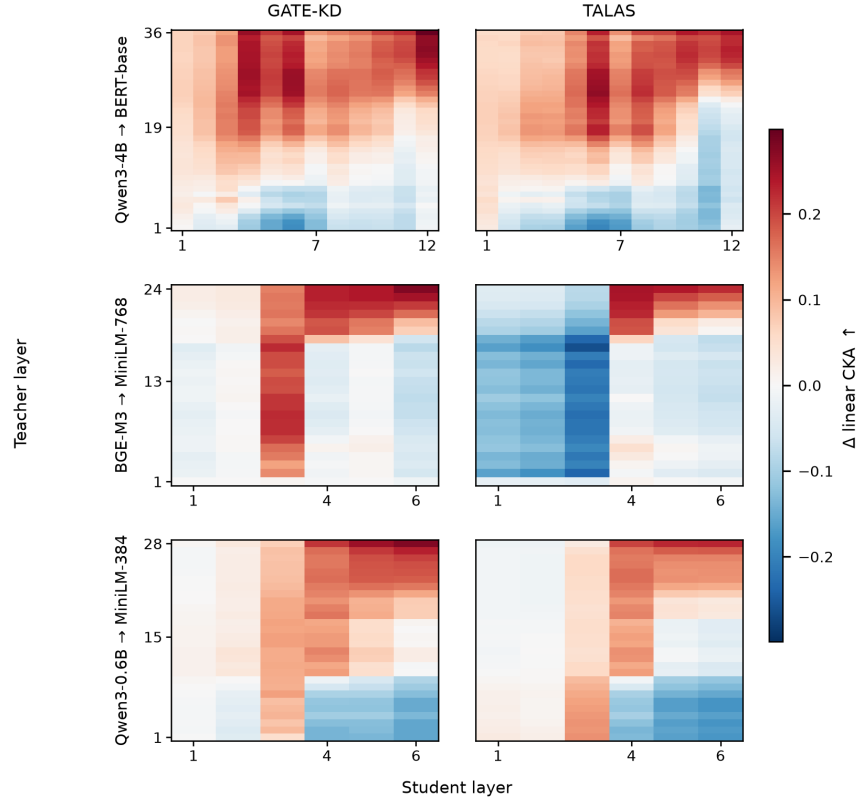


Figure 3: Change in layerwise linear CKA relative to the pretrained student. Red and blue denote increased and decreased similarity, respectively.

The changes spread over several later student layers rather than one teacher–student pair, and some layer pairs decrease, so final-embedding supervision co-occurs with changes in later layers without uniformly stronger alignment at every depth. CKA is descriptive: it does not establish that the observed changes cause downstream improvements or that endpoint supervision mechanistically substitutes for intermediate-layer distillation. It is also invariant to orthogonal transformations and isotropic scaling, so higher CKA measures representational similarity rather than transferred task information; in particular it is blind to the orthogonal gauge itself, so nothing in these figures bears on the coordinate-alignment claims of Section 4.3.2. The figures report seed means only; we have not measured seed-level variance of the CKA matrices, included a fixed-PCA control without a fitted gauge, or tested whether per-layer CKA changes correlate with downstream differences. We therefore use these probes to document that endpoint supervision co-occurs with changes in later layers, and for nothing stronger.